



A comprehensive in silico genome-wide identification and characterization of SQUAMOSA promoter binding protein (SBP) gene family in *Musa acuminata*

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ABSTRACT

One of the largest and most significant transcription factor gene families in plants is the SQUAMOSA promoter binding protein (SBP) gene family and they perform critical regulatory roles in floral enhancement, fruit development, and stress resistance. The SBP protein family (also known as SPL) has not yet been thoroughly studied in the staple fruit crop, banana. A perennial monocot plant, banana is essential for ensuring food and nutrition security. This work detected 41 SBP genes in the banana species *Musa acuminata*. The MaSBPs were subsequently elucidated by investigating their gene structure, chromosomal position, RNA-Seq data, along with evolutionary connections with Arabidopsis and rice. Sequence alignment of MaSBPs revealed that all genes included a domain of two Zn finger motifs (CCCH and CCHC motifs) with an overlapping nuclear localization signal region. The conserved motifs sequence in the inferred MaSBP proteins were quite comparable. According to findings, the time frame of divergence for duplicated MaSBP gene pairs ranged from 42.39 to 109.11 million years and the dicot Arabidopsis and monocotyledonous plant banana diverged before the division of banana and monocot rice. Moreover, cis-acting element and GO annotation analysis exhibited possible biological activities of MaSBPs in flower development, phytohormone regulation, and stress tolerance. RNA-Seq expression profiling exhibited that genes MaSBP-3, MaSBP-20, MaSBP-37, MaSBP-40 were more expressed during floral and fruit development stage. The foundation for additional investigation of SBP protein sequences in other plants can be laid out by this study, which will shed light on some of their crucial biological functions.

1. Introduction

Proteins called transcription factors (TFs) attach to DNA regulatory regions, usually located 5' upstream of target genes, to control the rate of gene transcription. They can either promote or inhibit the transcription of several target genes^{1,2}. Various biological processes rely on precise gene expression control and gene transcription is a crucial part of these systems³. Although TFs are shared by all eukaryotes, several TF families are unique to plants⁴.

The SQUAMOSA promoter binding protein (SBP) gene family is a group of genes exclusive to plants that encode zinc finger proteins. Each sequence in this family features a highly conserved DNA-binding domain consisting of approximately 80 amino acid residues and two Zinc-finger motifs (ZnF) (CCCH and CCHC types)⁵. Additionally, a conserved domain structural study confirmed that a nuclear localization

signal (NLS) slightly overlapped with the zinc-finger motif region at the C-terminus of SBP⁶. SBP genes are recognized to be involved in the transcriptional control of different biological and molecular processes linked to floral induction, fruit development, and to some extent they can govern diverse plant adaptations to stresses⁷. SBP gene-derived proteins that interact with the promoter region of SQUAMOSA, a floral meristem identifying gene, led to their discovery in *Antirrhinum majus*⁸. Subsequently, similar genes in Arabidopsis, referred to as SBP-like (SPL) genes, were identified⁹.

In the model plant *Arabidopsis thaliana*, 16 SBP-box genes identified and renamed SPL1 to SPL16¹⁰. SPL3 was detected to be substantially expressed within the floral meristem, leaf, vegetative, and inflorescence apex¹¹. SPL3, SPL4, and SPL5 were expressed significantly more during the long-day floral initiation process¹². SPL8 was also shown to have an impact on pollen sac formation¹³. One of the most important SBP-box

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Table 1Gene details and physicochemical properties of the 41 *MaSBP* genes in *M. acuminata*.

Transcript ID	Gene Name	Chr	CDS (bp)	A.A Length	Molecular Weight (kDa)	Aliphatic Index	pI	GRAVY	No of Intron/ Exon	E-value
GSMUA_Achr3G05360_001	<i>MaSBP-1</i>	3	1395	465	50694.82	63.32	9.13	−0.581	03:04	6.6e-31
GSMUA_Achr3G27750_001	<i>MaSBP-2</i>	3	1410	470	51391.34	64.33	8.63	−0.626	03:04	6.8e-32
GSMUA_Achr3G24520_001	<i>MaSBP-3</i>	3	2886	962	105940.51	83.59	5.32	−0.277	13:14	4.6e-30
GSMUA_Achr3G23780_001	<i>MaSBP-4</i>	3	2172	724	81061.62	81.31	5.91	−0.278	08:09	1.3e-24
GSMUA_Achr3G11170_001	<i>MaSBP-5</i>	3	1167	389	41307.88	53.4	9.18	−0.522	05:06	6.2e-28
GSMUA_Achr3G09190_001	<i>MaSBP-6</i>	3	2424	808	91977.54	82.61	7.01	−0.296	10:11	5.3e-27
GSMUA_Achr4G26940_001	<i>MaSBP-7</i>	4	1044	348	38175.18	70.95	9.02	−0.409	02:03	3.4e-31
GSMUA_Achr4G05720_001	<i>MaSBP-8</i>	4	1041	347	38500.05	55.09	9.34	−0.713	04:05	8.1e-32
GSMUA_Achr4G01450_001	<i>MaSBP-9</i>	4	1074	358	38204.66	58.49	8.9	−0.471	04:05	5.5e-18
GSMUA_Achr4G24610_001	<i>MaSBP-10</i>	4	987	329	35664.27	59.54	9.19	−0.399	01:02	1.0e-29
GSMUA_Achr4G12470_001	<i>MaSBP-11</i>	4	1053	351	39282.06	60.29	9	−0.66	05:06	6.7e-31
GSMUA_Achr4G16050_001	<i>MaSBP-12</i>	4	1083	361	39255.7	62.06	8.2	−0.581	03:04	2.5e-30
GSMUA_Achr7G02020_001	<i>MaSBP-13</i>	7	1041	347	37115.73	58.06	8.86	−0.488	03:04	1.8e-32
GSMUA_Achr7G22940_001	<i>MaSBP-14</i>	7	960	320	34864.26	69.37	8.86	−0.422	01:02	1.1e-30
GSMUA_Achr7G08750_001	<i>MaSBP-15</i>	7	1065	355	39105.8	62.4	7.32	−0.579	02:03	1.6e-31
GSMUA_Achr7G20620_001	<i>MaSBP-16</i>	7	999	333	36431.52	57.95	9.3	−0.892	04:05	2.8e-30
GSMUA_Achr7G20510_001	<i>MaSBP-17</i>	7	1026	342	36746.86	55.57	9.38	−0.593	03:04	7.6e-32
GSMUA_Achr2G11940_001	<i>MaSBP-18</i>	2	957	319	34276.41	59.06	8.56	−0.486	01:02	6.6e-30
GSMUA_Achr2G06150_001	<i>MaSBP-19</i>	2	1182	394	42562.47	63.82	6.5	−0.48	02:03	1.3e-30
GSMUA_Achr2G01820_001	<i>MaSBP-20</i>	2	3057	1019	113101.06	84.46	6.06	−0.274	12:13	4.2e-30
GSMUA_Achr9G14160_001	<i>MaSBP-21</i>	9	1377	459	50116.07	65.61	9.14	−0.562	03:04	4.9e-31
GSMUA_Achr9G22260_001	<i>MaSBP-22</i>	9	1032	344	36833.37	62.59	8.68	−0.459	02:03	1.8e-30
GSMUA_Achr9G23140_001	<i>MaSBP-23</i>	9	1251	417	46186.39	59.64	6.99	−0.695	04:05	1.6e-30
GSMUA_Achr9G27660_001	<i>MaSBP-24</i>	9	1110	370	39363.99	56.07	8.48	−0.518	02:03	2.1e-32
GSMUA_Achr9G29330_001	<i>MaSBP-25</i>	9	1149	383	41738.65	59.74	8.89	−0.655	02:03	4.2e-32
GSMUA_Achr9G17440_001	<i>MaSBP-26</i>	9	2928	976	110118.35	85.61	6.31	−0.202	13:14	2.5e-27
GSMUA_Achr9G01380_001	<i>MaSBP-27</i>	9	1122	374	40792.21	69.57	8.46	−0.406	03:04	5.3e-31
GSMUA_Achr8G23820_001	<i>MaSBP-28</i>	8	1584	528	57777.44	64	9.22	−0.539	05:06	2.5e-30
GSMUA_Achr8G25550_001	<i>MaSBP-29</i>	8	1035	345	36962.94	48.58	8.41	−0.624	03:04	8.0e-33
GSMUA_Achr11G19580_001	<i>MaSBP-30</i>	11	1248	416	46400.77	72.55	9.33	−0.387	03:04	1.7e-30
GSMUA_Achr6G06030_001	<i>MaSBP-31</i>	6	1110	370	39952.84	67.64	7.65	−0.451	02:03	2.4e-32
GSMUA_Achr6G36010_001	<i>MaSBP-32</i>	6	1317	439	48339.04	65.68	8.69	−0.57	03:04	1.8e-31
GSMUA_Achr6G22280_001	<i>MaSBP-33</i>	6	1236	412	43815.27	62.75	8.54	−0.422	03:04	4.5e-33
GSMUA_Achr10G02970_001	<i>MaSBP-34</i>	10	1515	505	55547.33	72.02	8.84	−0.479	04:05	3.2e-31
GSMUA_Achr10G23280_001	<i>MaSBP-35</i>	10	1464	488	53926.51	67.64	9.17	−0.591	04:05	8.9e-30
GSMUA_Achr5G25090_001	<i>MaSBP-36</i>	5	1242	414	45121.85	68.21	8.31	−0.445	04:05	1.7e-30
GSMUA_Achr5G22250_001	<i>MaSBP-37</i>	5	2934	978	108920.3	75.16	8.43	−0.474	11:12	1.2e-29
GSMUA_Achr5G22630_001	<i>MaSBP-38</i>	5	1365	455	49890.86	61.63	8.62	−0.507	03:04	1.6e-31
GSMUA_Achr5G21990_001	<i>MaSBP-39</i>	5	1302	434	48331.1	62.15	9.17	−0.75	04:05	2.0e-31
GSMUA_Achr5G27280_001	<i>MaSBP-40</i>	5	3198	1066	118386.34	77.1	6.73	−0.471	13:14	2.9e-28
GSMUA_Achr5G26880_001	<i>MaSBP-41</i>	5	2334	778	88020.39	74.8	7.39	−0.421	10:11	9.2e-25

*Chr: Chromosome, A.A: Amino acid, pI: Theoretical isoelectric point, GRAVY: Grand average of hydropathicity.

genes, *SPL14*, has recently been found to confer tolerance to programmed cell death (PCD) inducing fungal toxin fumonisins B1 (FB1)¹⁴. The complex role of *AtSPL2/10/11/13* aided in regulation of the transition from vegetative to reproductive phase¹⁵.

Rice is a major crop that serves as a model monocotyledon organism for molecular and genetic research¹⁶. In rice, *OsSPL10* was shown to be activated only to govern trichome growth and salt tolerance in the inflorescence stem region¹⁷; *OsSPL14* was involved in both yield and immunity¹⁸, and *OsSPL2/17/18* influenced the amount of productive tillers and plant height¹⁹. In the last few years, in-silico analysis of *SBP* (*SPL*) gene family has been undertaken in various plants, including apple²⁰, grape²¹, castor bean²², melon²³, cotton²⁴, peanuts²⁵, chrysanthemum⁷, and pepper²⁶.

Banana production ranks fourth in agricultural production, with an estimated 106 million tons²⁷ and provides livelihoods of millions of small-scale agricultural producers. This perennial monocot plant is an excellent source of minerals, vitamins, fiber, and cellulose (utilized in paper and textiles). Most banana cultivars are derived from *M. acuminata* (2n = 2 × = 22, A genome), which occasionally hybridizes with *Musa balbisiana* (2n = 2 × = 22, B genome)²⁸. The limited genetic diversity makes banana plants very prone to pests and diseases, which results in the heavy application of insecticides and other pesticides to the crops. As a result, genetic engineering-based agricultural enhancement has a lot of potential. Candidate gene targets must be found for this purpose²⁹. The discovery and characterization of different gene families

has become extremely important to banana plant breeders and scientists, given the relevance of resistance to environmental risks and the fact that the entire banana genome is now available. This research is the first to utilize bioinformatics methods for a comprehensive genome-wide study of the *SBP* (*SPL*) gene family in banana.

The study found forty-one *MaSBP* sequences and investigated their molecular features using the most recent database of banana genome A. Moreover, the foundation for explaining the evolution of the *MaSBP* genes was acquired by the analysis of exon–intron structure, phylogenetic distribution, and anticipated motif composition. The distribution of genes over a wider range of molecular processes and cell organelles was identified by combining gene ontology analysis with subcellular localization prediction. The analysis of *cis*-acting elements in upstream regulatory regions uncovered the presence of specific critical motifs associated with the functions of *MaSBPs* in response to biotic and abiotic stress. Additionally, this plant-specific transcription factor family was also examined in dicot and monocot model species, namely *Arabidopsis* and rice, with *M. acuminata* (Banana), utilizing the complete-length *SBP* protein sequences to create a neighbor-joining phylogenetic tree. An analysis of the nonsynonymous and synonymous substitution rates (Ka/Ks) for paralogs was conducted to explore the factors influencing gene evolution. Furthermore, the curated RNA-Seq data of *M. acuminata* *SBP* gene family explained how these genes upregulated and downregulated in response to certain conditions.

The study investigates the *SBP* gene family in bananas, highlighting

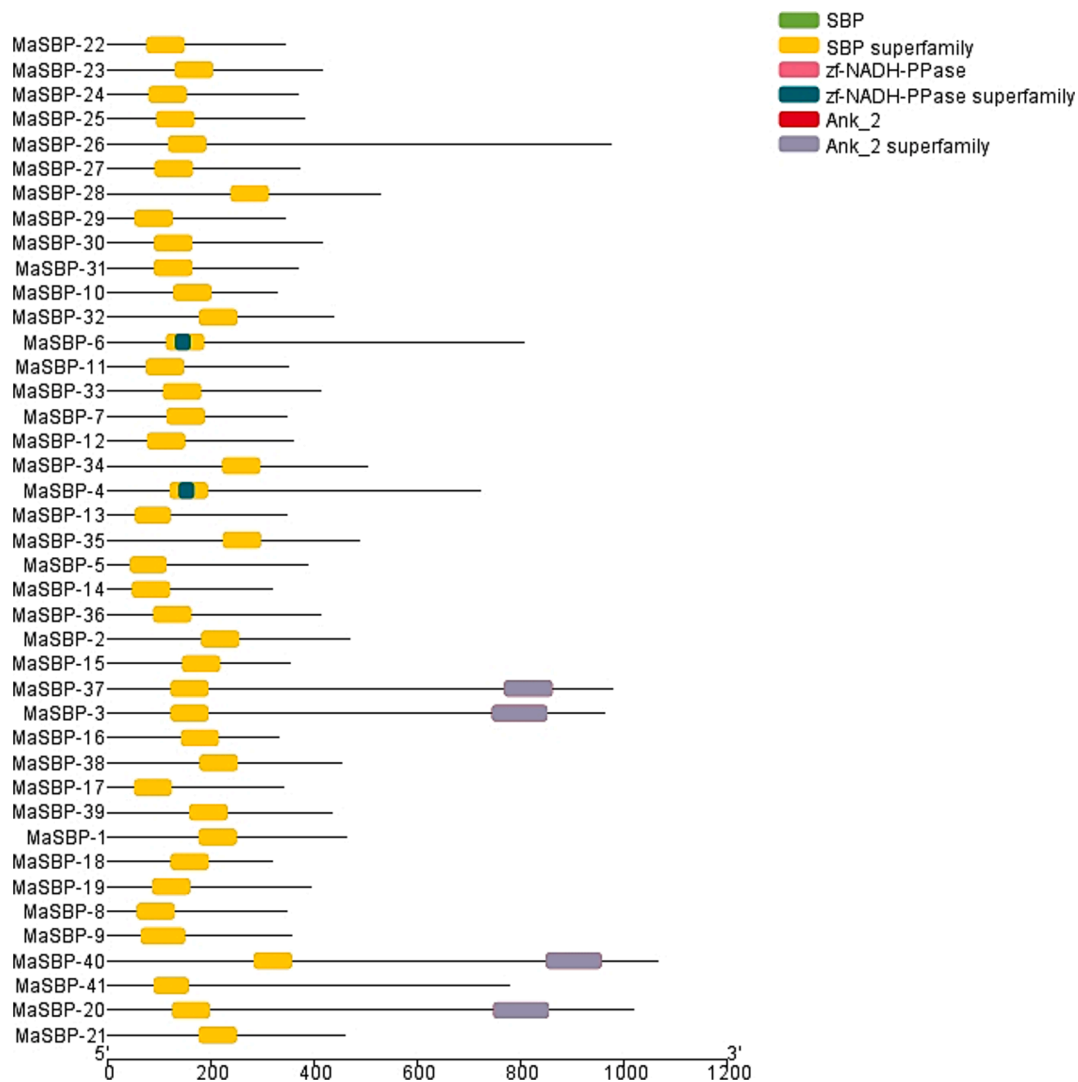


Fig. 1. Conserved domains of *MaSBPs*. Different domains are represented in different colors.

its functional characterization and regulatory roles in critical processes such as fruit ripening and responses to environmental stressors. The findings suggest that *SBP* genes are integral to understanding how bananas adapt to stress and regulate maturation, providing insights that could enhance agricultural practices and improve crop resilience. Additionally, it is possible that this gene family would be a great prospect for banana genomic research. This information would facilitate the identification of genes linked to commercially valuable traits in *Musa* and could be utilized in banana breeding programs to enhance desirable characteristics, potentially improving production and quality.

2. Materials and methods

2.1. Identification of *SBP* gene family sequences in *M. acuminata*

Using the *A. thaliana* SQUAMOSA promoter-binding protein-like (*SBP* domain) transcription factor family protein sequence (NCBI Reference Sequence: NP_851161.1), the online database Phytozome v13.0³⁰ was employed to identify the *SBP* genes within the banana genome (*M. acuminata*)³¹. The inappropriate predicted sequences were manually eliminated³², and the existence of *SBP* specific domains in non-redundant *SBP* protein sequences was validated using the web tool MOTIF Search (<https://www.genome.jp/tools/motif/>) and HMMER v.3.3.2 (<https://www.ebi.ac.uk/Tools/hmmer/>)³³. Finally, 41 protein

sequences were obtained, and were renamed *MaSBP-1* to *MaSBP-41*. NCBI Conserved Domain Search (<https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi>)³⁴ and TBtools algorithm “Visualize NCBI CDD Domain Pattern”³⁵ were used for domain detection. Phytozome v13.0 was used to collect the length of amino acids, chromosomal location, genomic sequences, CDS sequences and 1.5 kbp upstream promoter regions of the validated genes from the banana genome. ExPASy tools (<https://web.expasy.org/protparam/>) were used to compute the molecular weight, theoretical pI, aliphatic index, and grand average of hydropathicity (GRAVY) utilizing standard parameters and the protein sequence of the relevant gene³⁶. Moreover, the *SBP* gene sequences of *A. thaliana* (TAIR10) and *O. sativa* ssp. *japonica* cv. *Nipponbare* (*O. sativa* v7.0) were retrieved from phytozome and the potential candidates were identified using the same approach as mentioned above. The curated *SBP* genes of *A. thaliana* and *O. sativa* were renamed *AtSBP-1* to *AtSBP-15* and *OsSBP-1* to *OsSBP-16*, respectively.

2.2. Multiple sequence alignment (MSA) and phylogenetic analysis

Arabidopsis, rice, and banana *SBP* protein sequences alignment was performed using the MEGA-X (v11) MUSCLE program³⁷. The alignment was subsequently examined with the JalView-2.11.2.2 software³⁸. The phylogenetic analysis was conducted using the MEGA-X software package, employing the neighbor-joining (NJ) algorithm, poisson

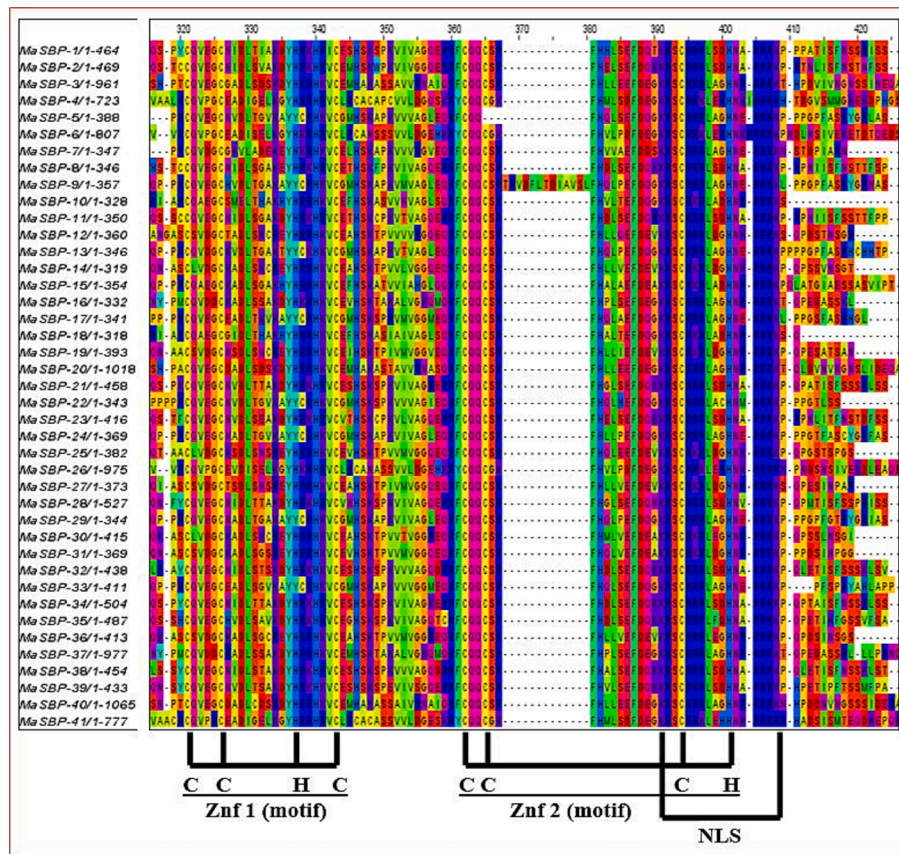


Fig. 2. Multiple sequence alignment (MSA) of *MaSBPs*. Two conserved zinc finger structures (CCHC and CCCH) and nuclear localization signal (NLS) are indicated.

correction, and a bootstrap test with 1,000 re-samplings to classify and depict the evolutionary relationships of *SBPs* identified from banana, rice and Arabidopsis.

2.3. Gene structure and conserved motifs

The Gene Structure Display Server (GSDS 2.0) (<https://gsds.gao-lab.org/>) was used to visualize the locations of exon–intron in the identified *MaSBP* genes by comparing each coding sequence to its respective genomic sequence³⁹. The MEME (Multiple Em for Motif Elicitation) tool (<https://meme-suite.org/meme/tools/meme>) of MEME suite software (version 5.4.1) was employed to analyze the motifs in each predicted *MaSBP* sequence⁴⁰. The TBtools application “Visualize MEME/MAST Motif Platform” was also used to create a *MaSBPs* conserved motifs graph³⁵ and the consensus sequences of discovered motifs were scanned against the PROSITE database with the ScanPROSITE program (<https://expasy.org/tools/scanprosite/>)¹⁶.

2.4. Subcellular localization and chromosomal location prediction

WoLF PSORT (<https://wolfpsort.hgc.jp/>) was utilized to predict subcellular localization⁴¹. Using the TBtools Heatmap program, a heatmap was created based on the WoLF PSORT result³⁵. An online tool called MG2C (MapGene2Chrom v.2.1) was employed to determine the chromosomal position (https://mg2c.iask.in/mg2c_v2.1/)⁴².

2.5. Analysis of cis-acting regulatory elements and gene ontology (GO)

The 1.5 kb upstream promoter region of each *MaSBP* gene (collected from phytozome database) was investigated using the Plant cis-acting Regulatory DNA Elements (Plant-CARE) tool (<https://bioinformatics.ps.b.ugent.be/webtools/plantcare/>)⁴³. A gene product’s molecular function,

physiological process, and cellular localization all are described by Gene Ontology (GO)⁴⁴, that can result in several GO terms being allocated to a single query sequence. *MaSBP* protein molecular functions were predicted using the PlantRegMap Go Term Enrichment tool (<https://plantregmap.gao-lab.org/go.php>)⁴⁵. The results were visualized via a graph, which was constructed using Origin software (version 2023) from OriginLab.

2.6. Paralogous gene pairs (duplicated) and K_a , K_s , K_a/K_s values calculation

The phylogenetic tree was used to derive the paralogs for *SBP* genes in *M. acuminata*¹⁶. The synonymous (K_s), non-synonymous (K_a) substitution rate and their ratio for each paralogous pair were calculated using the TBtools software “Simple K_a/K_s Calculation (NG)”³⁵. The approximate timing of the duplication events was estimated using the formula $T = K_s / 2\lambda \times 10^{-6}$ Mya (million years ago), [where, (λ) in Banana (*Musa* spp.) is 4.5×10^{-9}]⁴⁶. Using the TBtools Advance Circos platform, the link among *SBP* gene paralogs was visualized³⁵.

2.7. Synteny analysis (orthologous gene pairs)

The TBtools Synteny visualization plugin was utilized to establish synteny links between orthologous *SBP* genes in banana, Arabidopsis and rice⁴⁷. TBtools “One Step MCScanX” program identified which “syntenic genes” were employed for synteny analysis³¹. The orthologous of *SBP* genes situated on syntenic chromosomal blocks, were illustrated using the “Dual Synteny Plot” in TBtools³⁵.

2.8. Analysis of RNA sequencing data for *M. acuminata*

To gain insight into the expression of *SBP* genes in *M. acuminata*,

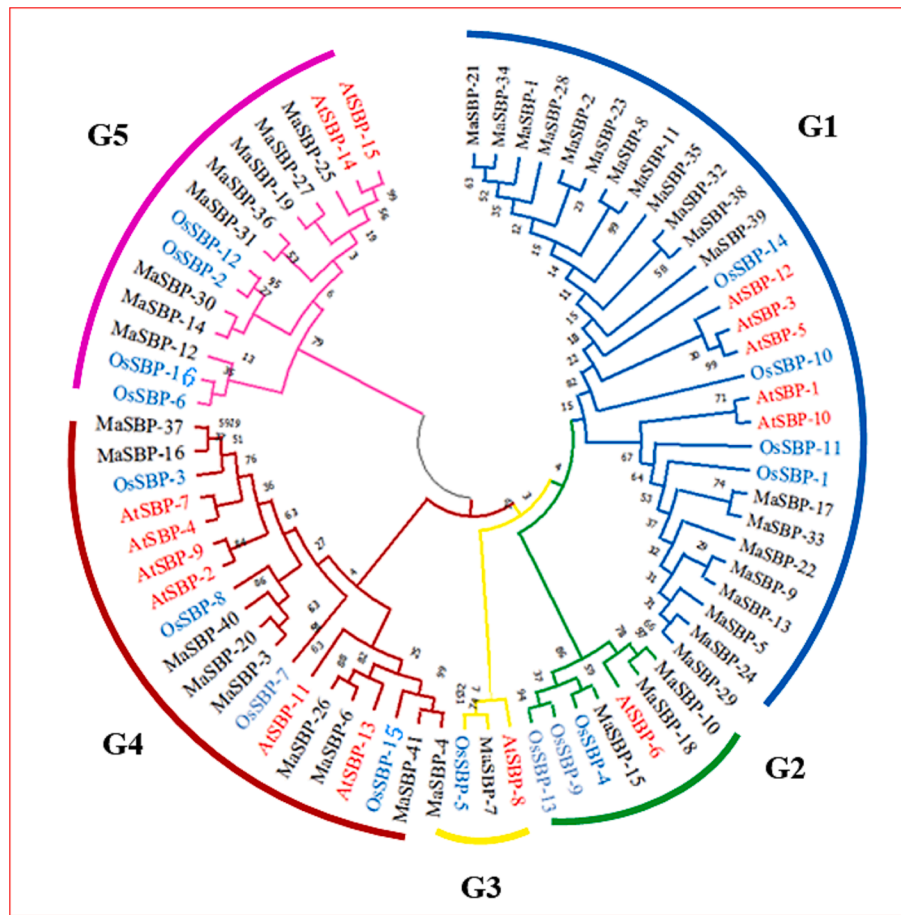


Fig. 3. SBP proteins un-rooted phylogenetic tree from *M. acuminata*, *O. sativa*, and *A. thaliana*. MEGA11 software constructed the tree through Neighbour-joining (NJ) algorithm. The numbers above the branches represent the bootstrap value. Different colors represent different plants. *M. acuminata* is represented by black, *A. thaliana* by red, and *O. sativa* by blue. The tree was categorised into five groups. Different colors are used to emphasize different category (group 1: blue, group 2: green, group 3: yellow, group 4: maroon, and group 5: pink).

RNA-Seq data was retrieved from the Expression Atlas database (<https://www.ebi.ac.uk/gxa/home/>)⁴⁸. The Atlas database offered details on the baseline expression and tissue-specific differential expression of all *MaSBP* genes. The TBtools Heatmap program was employed to create a baseline expression heatmap, using the exon model Fragments Per Kilobase of transcript per Million mapped reads (FPKM) value³⁵. Another differential expression graph was constructed via Origin software (version 2023) from OriginLab based on the log₂-fold change (Log₂FC) scores of a few specific *MaSBP* genes, that are currently available in Atlas database.

3. Results

3.1. SBP gene identification and physicochemical properties

A set of procedures was used to identify the SBP gene family members in *M. acuminata*. The HMM profile of the conserved SBP domain (PF03110) was employed as a blast query to search against the available banana (*M. acuminata*) genome database to locate SBP genes in banana (Phytozome v13). The sequences that were too short were removed. After conducting a MOTIF Search and analyzing the SBP domain using the HMMER tool, a total of 41 putative SBP members were detected and denoted *MaSBP-1* to *MaSBP-41*. HMMER also calculated the E-value (expectation value) of individual protein sequence SBP domain or motif and the values were provided in Table 1. This E-value indicates the significance of sequences in the whole database search. In general, a lower E-value ($\leq 10^{-5}$), is equivalent to P-value (probability value).

Every selective *MaSBP* sequence demonstrated lower E-value than 10^{-5} , suggesting a highly significant match. Furthermore, all *MaSBPs* domain were analyzed by NCBI and TBtools and the constructed graph indicated SBP domain presence in every protein sequence (Fig. 1). Selected protein's gene accessions, chromosomal number, the coding sequences (CDS) lengths, amino acid lengths etc. related information were collected from phytozome database and were arranged in Table 1. All the curated *MaSBP* genes were found on total ten chromosomes of *M. acuminata*. The lowest numbers of *MaSBP* genes were located on chromosome 11 (one), whereas the maximum numbers of *MaSBP* were identified on chromosome 9 (seven). *MaSBP* amino acid lengths ranged from 319 (*MaSBP-18*) to 1066 (*MaSBP-40*). The *MaSBP* genes' CDS length ranged from 957 (*MaSBP-18*) to 3198 (*MaSBP-40*). The ExPASy ProtParam program was used to estimate several molecular features of *MaSBPs*, including molecular weight, theoretical pI, aliphatic index, and grand average of hydropathicity (GRAVY). According to Table 1, the molecular weight of *MaSBPs* ranged from 34276.41 kDa (*MaSBP-18*) to 118386.34 kDa (*MaSBP-40*), and the pI of *MaSBP* genes varied from 5.32 (*MaSBP-3*) to 9.38 (*MaSBP-17*). The pI (isoelectric point) is the pH at which a protein has no net charge. The aliphatic index of a protein refers to the volume occupied by aliphatic side chains (alanine, valine, isoleucine, and leucine). The aliphatic index of these proteins ranged from 48.58 (*MaSBP-29*) to 85.61 (*MaSBP-26*). The grand average of hydropathicity (GRAVY) score was below zero for all *MaSBP* sequences (Table 1) indicating hydrophilic nature of those proteins.

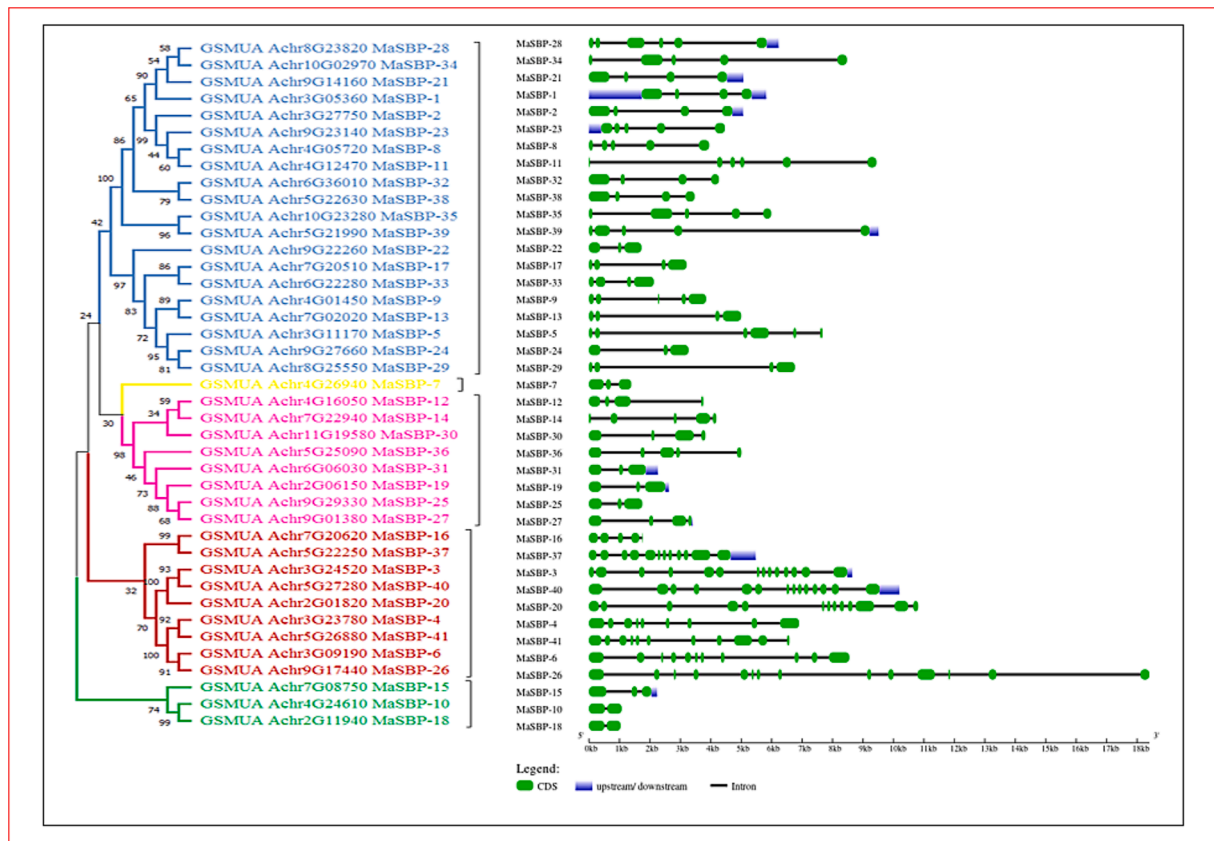


Fig. 4. *MaSBP* genes structure. Evolutionary tree was constructed through the Neighbour-Joining (NJ) approach. Exons (CDS) are depicted in green, introns are shown as grey lines, and untranslated regions (UTRs) are indicated in blue.

3.2. Multiple sequence alignment and phylogenetic relationship

According to the findings of the MUSCLE program in MEGA-X software, each SBP protein had an entire SBP domain with two ZnF motifs and one nuclear localization signal region (NLS), as demonstrated in Fig. 2. The first zinc finger motif was of the C2HC type, while the second was of the C3H type. SBP domain-containing proteins from three plant genomes (*M. acuminata*, *A. thaliana*, and *O. sativa ssp. japonica*) were incorporated into the data analysis to explore the phylogenetic connections and evolutionary history of *MaSBPs*. The phytozome database was used to find the SBP protein sequences of Arabidopsis (15) and cultivated rice (16). A phylogenetic tree of proteins containing the SBP domain was created using amino acid sequences and bootstrap values from 1,000 repetitions (Fig. 3) through the neighbor-joining (NJ) method to assess the relationships among *MaSBPs*. The resulting evolutionary tree classified the SBP proteins into five groups, each containing at least one *MaSBP* sequence. There were 20 *MaSBP* genes in the biggest group 1. The smallest group 3 only had one member. Other groups, such as groups 2, 4, and 5, had 3, 9, and 8 *MaSBP* members, respectively.

3.3. Genes structure and conserved motifs assessment

The GSDS software was used to match full-length cDNA sequences of SBP genes to their respective genomic sequences. Forty-one banana SBP genes patterns were studied with their rectangle phylogenetic tree. In terms of gene structure analysis, these genes were grouped into five groups, as shown in Fig. 4, in accordance with the evolutionary tree of *MaSBP* proteins. Not all genes had the same number of exons and introns (Fig. 4; Table 1). The estimated number of exons ranged from 2 to 14, while the number of introns ranged from 1 to 13. *MaSBP-3* and *MaSBP-*

26 contained highest number of exons and introns. Twelve genes (*MaSBP-1*, *MaSBP-2*, *MaSBP-3*, *MaSBP-15*, *MaSBP-19*, *MaSBP-21*, *MaSBP-23*, *MaSBP-28*, *MaSBP-31*, *MaSBP-37*, *MaSBP-39*, and *MaSBP-40*) possessed untranslated regions (UTRs) (Fig. 4). The MEME program was employed to search for motifs in all *MaSBP* genes, using the default settings but with the maximum number of motifs to identify set to 10. The ScanPROSITE program identified two out of ten motifs as part of the SBP gene family. Among the ten motifs discovered, two (motif 1 and motif 2) were highly conserved and detected in nearly all SBP genes (Fig. 5B). The lengths of the motifs varied from 6 to 50 amino acids. TBtools result was identical to the MEME result, in which 41 *MaSBP* genes included motif 1 and 2, both of which contained the SBP domain among all motifs (Fig. 5C). The E-values of the two most prominent conserved motifs found in all *MaSBPs* were 4.4E-1071 (Motif-1) and 2.3E-1316 (Motif-2) (Table 2). These two motifs comprised C3H and C2HC zinc-finger structures, as well as a bipartite nuclear localization signal (NLS) of the conserved SBP domain (Fig. 5D).

3.4. Subcellular distribution and chromosomal location

The subcellular localization prediction of the identified *MaSBP* proteins anticipated that most of the proteins reside within the nucleus (Fig. 6). Prediction results were given on cytosol (cyto), endoplasmic reticulum (ER), extracellular (extra), golgi apparatus (golgi), chloroplast (chlo), mitochondria (mito), nuclear (nucl), peroxisome (pero), plastid (plas), and vacuole (vacu). For localization prediction, a scoring system was utilized. The WoLF PSORT program produced the result by comparing *MaSBP* protein to its 14 closest neighbours. Except for *MaSBP-3* (golgi), *MaSBP-6* (plasmamembrane), and *MaSBP-26* (plasmamembrane), practically all these neighbours were predominantly found in the nucleus compartment (Fig. 6). Analysis of chromosomal

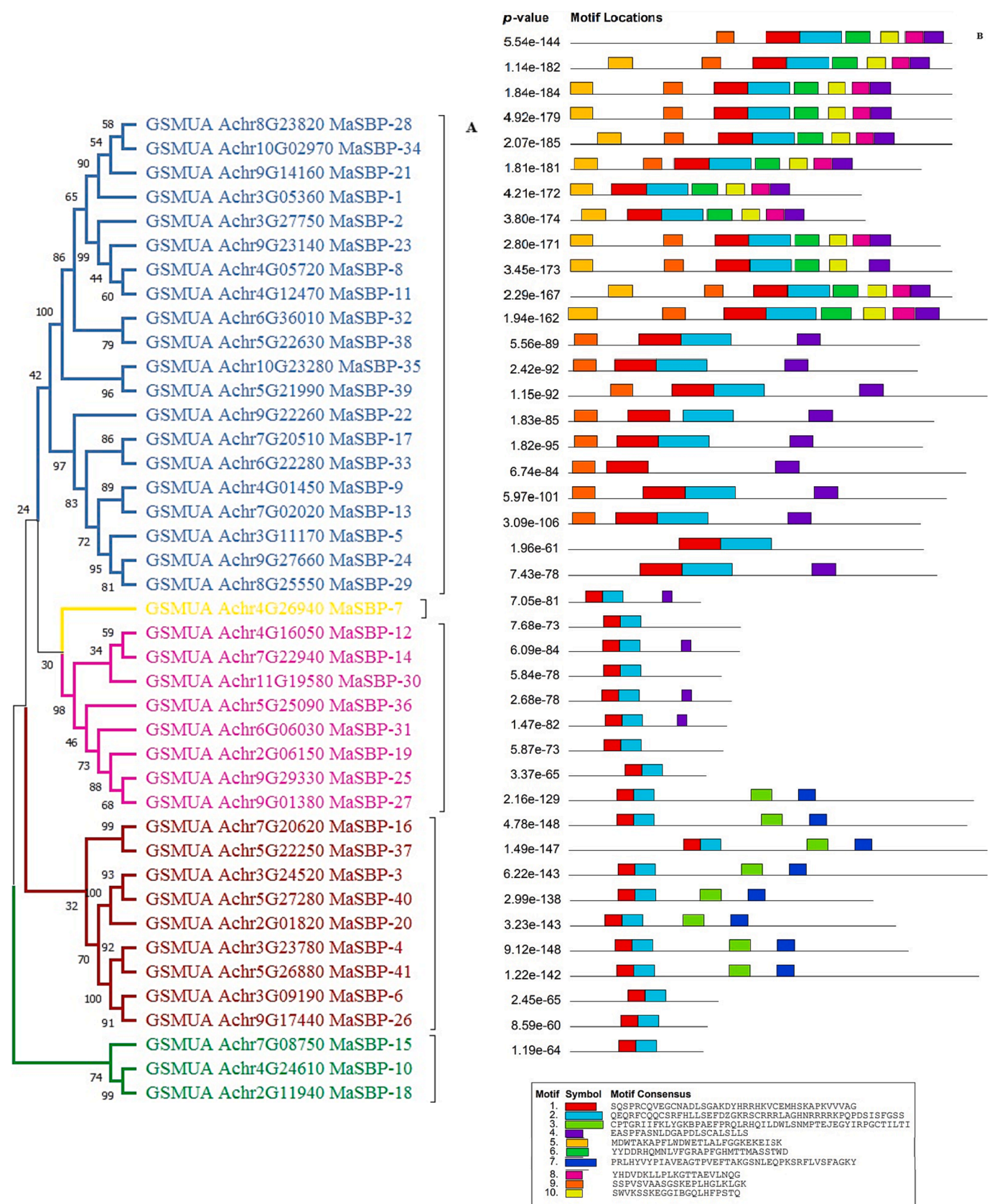


Fig. 5. Conserved motifs of *MaSBP* proteins. (A) Neighbour joining tree constructed for *MaSBPs*. (B) Conserved motifs by MEME suite. Various colored boxes represent distinct motifs. (C) Conserved motifs of *MaSBPs* by TBtools. (D) Sequence logos (motif 1 and 2) with the *MaSBP* genes' conserved *SBP* domain comprising the two Znf and NLS motifs, obtained from the MEME suite.

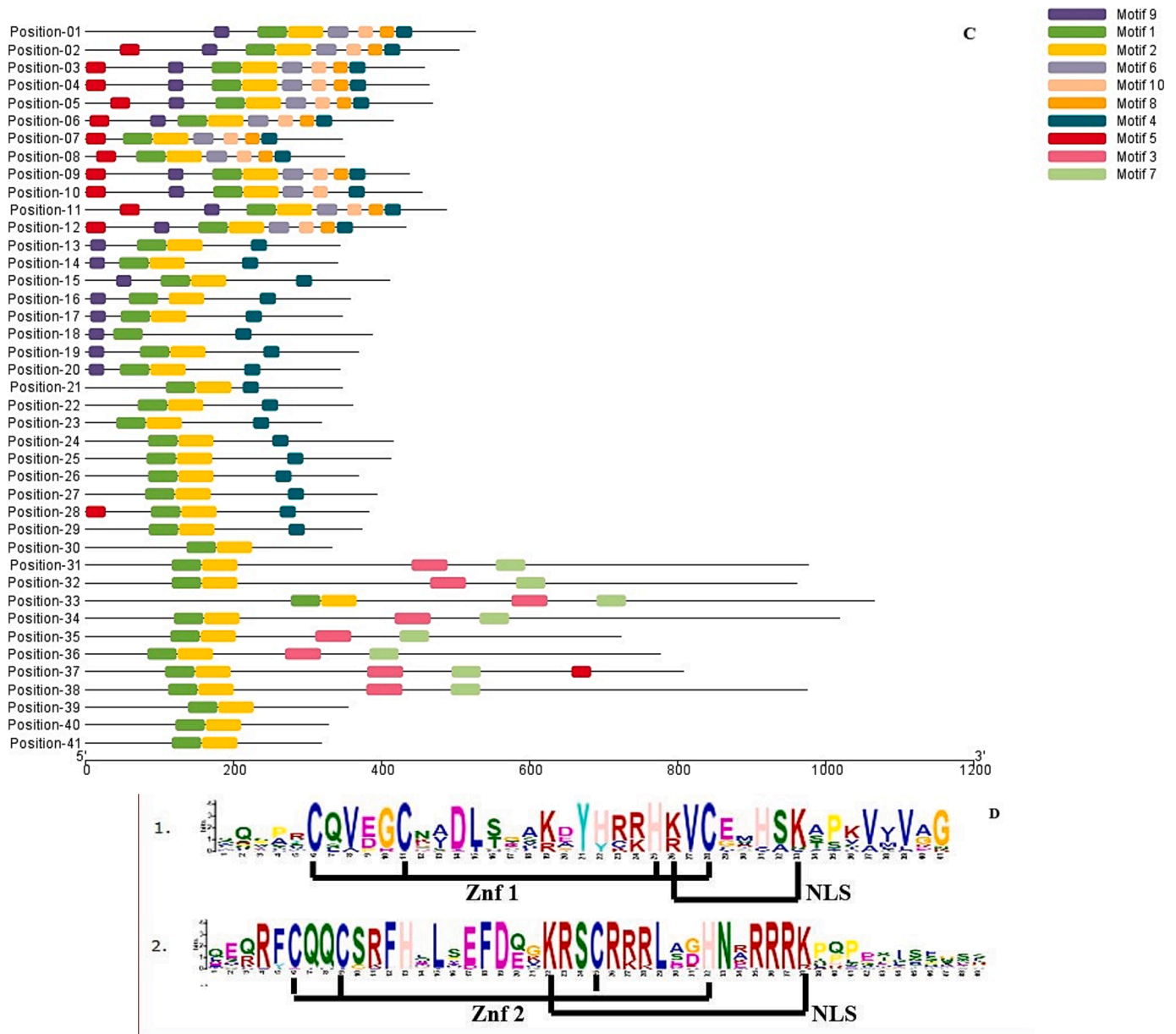


Fig. 5. (continued).

Table 2
MaSBPs MEME motif sequences and lengths.

Motif ID	Conserved Sequence	E- value	Width	Description (Domain)
1	SQSPRCQVEGCNADLSGAKDYHRRHKVCEMHSAKPKVVVAG	4.4E-1071	41	SBP
2	QEQRFCQQCSRFLHLLSEFDZGKRSCRRLAGHNRRRKQPQDSISFGSS	2.3E-1316	49	SBP
3	CPTGRIIFKLYGKBPAEFPRQLRHQILDWLSNMPTJEGYIRPGCTILTI	2.00E-205	50	Unknown
4	EASPFASNLGDGAPDLSICALSLLS	3.00E-141	23	Unknown
5	MDWTAKAPFLWDWETLALFGGKEKEISK	5.10E-123	28	Unknown
6	YYDDRHQMNLVFGRAFGHMTTMASTWD	1.90E-119	29	Unknown
7	PRLHYVYPIAVEAGTPVEFTAKGSNLEQPKSRFLVSFAGKY	1.30E-97	41	Unknown
8	YHDVDKLLPLKGTTAEVNLQG	4.10E-75	21	Unknown
9	SSPVSAASGSKEPLHGLKLKG	6.30E-62	22	Unknown
10	SWVKSSKEGGIGBQLHFPSTQ	3.10E-54	21	Unknown

location indicated that 41 *MaSBPs* were irregularly distributed across the chromosomes. Approximate positions of *MaSBP* genes were investigated on chromosomes to see how they were distributed across the genome. Only ten out of the twelve *M. acuminata* chromosomes (chr 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) contain all forty-one *SBP* genes. Chromosome 1

and the unidentified chromosome do not have any genes. The maximum numbers of *MaSBP* genes (seven) were found on chromosome 9, while the lowest counts of *MaSBP* genes were found on chromosome 11 (one) (Fig. 7). On chromosome 9, *MaSBP*-21, *MaSBP*-22, *MaSBP*-23, *MaSBP*-24, *MaSBP*-25, *MaSBP*-26, and *MaSBP*-27 were found. The *MaSBP*-30

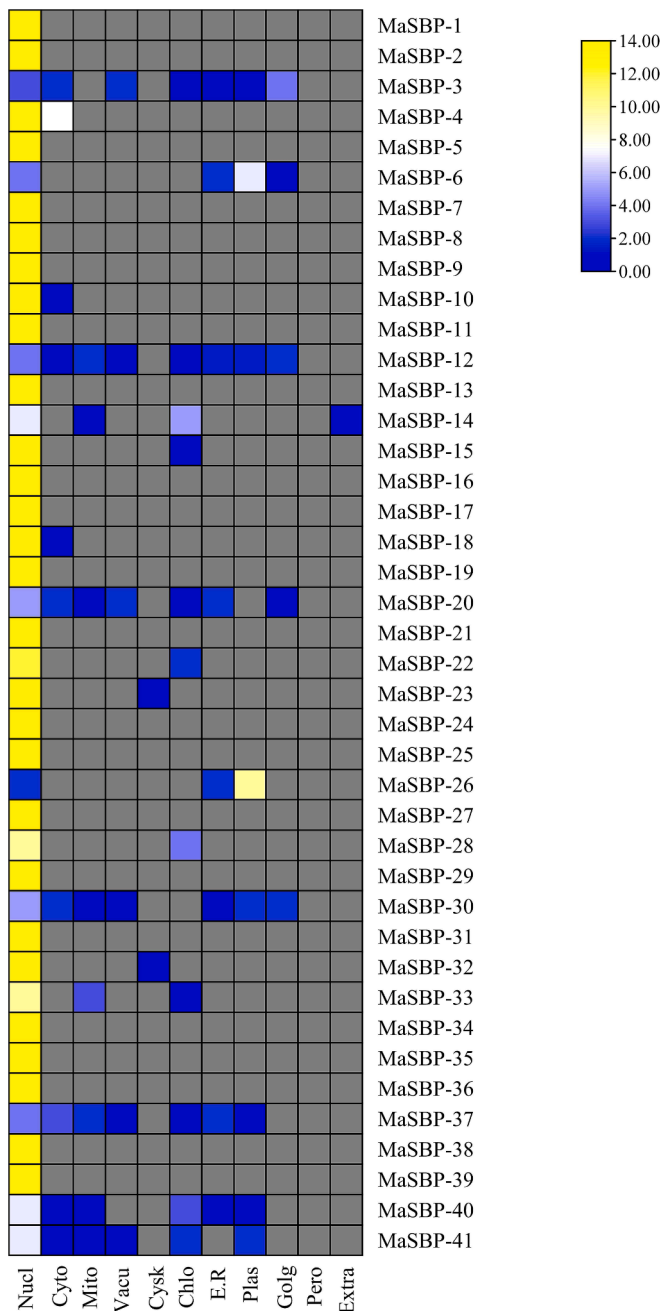


Fig. 6. Heatmap of *MaSBPs* subcellular localization through WoLF PSORT. Different colour indicates certainty level of protein existence.

gene was detected on chromosome 11.

3.5. Cis-acting regulatory elements and GO annotation

Six groups of *cis*-acting elements were detected in *MaSBPs*: Light responsive, Hormone related elements, Environmental stress related elements, Site binding, Development related elements and Transcription related elements (Fig. 8; Supplementary Table 1). All 41 *MaSBPs* contained transcription-related *cis*-elements, such as the TATA-box (core promoter element) and CAAT-box. The methyl jasmonate (MeJA)-responsive element and the abscisic acid (ABA)-responsive element are two prominent hormone-related *cis*-elements (ABRE; CGTCA-motif) that were found in *MaSBP* genes. Environmental stress-related *cis*-acting elements such as drought response element (MYB-binding site [MBS]), low-temperature response element (LTR), and wounding and pathogen

response element (W box) were also evaluated. Twenty-two and twenty-nine *MaSBP* contained MBS and ARE elements respectively. Meristem-specific and endosperm expression regulatory elements were among the development-related *cis*-elements (CAT-box; GCN4 motif). The existence of these CREs highlights their crucial role in plant growth and flower development, which are vital for regulating gene transcription in response to specific stresses or hormones in particular environmental conditions. GO enrichment analysis revealed the probable molecular activity, cellular positioning, and physiological action of the putative *SBPs* in *M. acuminata* based on systemic GO annotations. All the GO terms with their GO IDs and counts were organized in a tabular form (Supplementary Table 2). Practically all *MaSBPs* were identified with the GO term “GO:0003677”, showing their DNA binding activities (Fig. 9). Additional molecular functions associated with the 41 *MaSBP* genes included organic cyclic compound binding (GO:0097159), heterocyclic compound binding (GO:1901363), and nucleic acid binding (GO:0003676) respectively. Furthermore, three *MaSBP* sequences (*MaSBP*-19, *MaSBP*-21, *MaSBP*-29) were anticipated to be involved in the developmental process of anther, floral organ, phyllome, shoot system, reproductive system, anatomical structure etc. All *MaSBP* were predicted to be located in various cellular components, including the nucleus (GO:0005634), intracellular organelles (GO:0043229), and cell parts (GO:0044464), among others.

3.6. Analysis of collinear and selection pressure

Paralogous gene pairs were identified from the evolutionary tree (Fig. 4) to study the connection of *SBP* genes within species. A total of 14 paralogous gene pairings found in *MaSBPs*. Furthermore, *MaSBP* gene pairs' *Ks* (synonymous substitution rate) and *Ka* (non-synonymous substitution rate) values were estimated. In recent years, the *Ka/Ks* ratio has gained popularity as a criterion in genome-wide gene family identification studies. This ratio provides valuable insight to examine the evolutionary pressure of genes producing *SBP* proteins to measure their selection pressure^{49,50}. A *Ka/Ks* ratio greater than 1 indicates positive selection during gene sequence evolution. In contrast, a *Ka/Ks* value less than 1 suggests that duplications have undergone purifying (negative) selection, while a *Ka/Ks* value of 1 signifies that the duplications have experienced neutral selection⁵¹. The divergence period for duplicated *MaSBP* gene pairs ranged from 42.39 to 109.11 million years (MYA), as detailed in Table 3. The *Ka/Ks* ratio for all duplicated *MaSBP* pairs was less than 1, suggesting that purifying selection took place following the duplication events. A circos plot was created to illustrate the occurrences of *MaSBP* gene duplications on the chromosomes of *M. acuminata* (Fig. 10).

3.7. Synteny and evolutionary analysis (orthologous gene pairs identification)

A synteny analysis was conducted among *M. acuminata*, *A. thaliana*, and *O. sativa* to gain a deeper understanding of the evolutionary mechanisms of the *SBP* gene family in banana. Between the genomes of *M. acuminata* and *A. thaliana*, two *SBP* orthologous pairs were identified (Fig. 11A). In *M. acuminata* and *O. sativa*, twenty-four orthologous pairings were detected (Fig. 11B). The transcript ID of all orthologous gene pairs with their respective chromosome is given in Supplementary Table 3. Multiple *MaSBP* genes have an evolutionary link with *OsSBPs*. *MaSBP*-*AtSBP* had less ortholog pairings than *MaSBP*-*OsSBP*, indicating that the divergence in common ancestor of dicot Arabidopsis and monocotyledonous plant banana happened before the division of banana and rice (monocot).

3.8. RNA-sequencing data investigation

The Expression Atlas comprises high-quality microarray and RNA-sequencing (RNA-Seq) data sourced from ArrayExpress⁵², along with

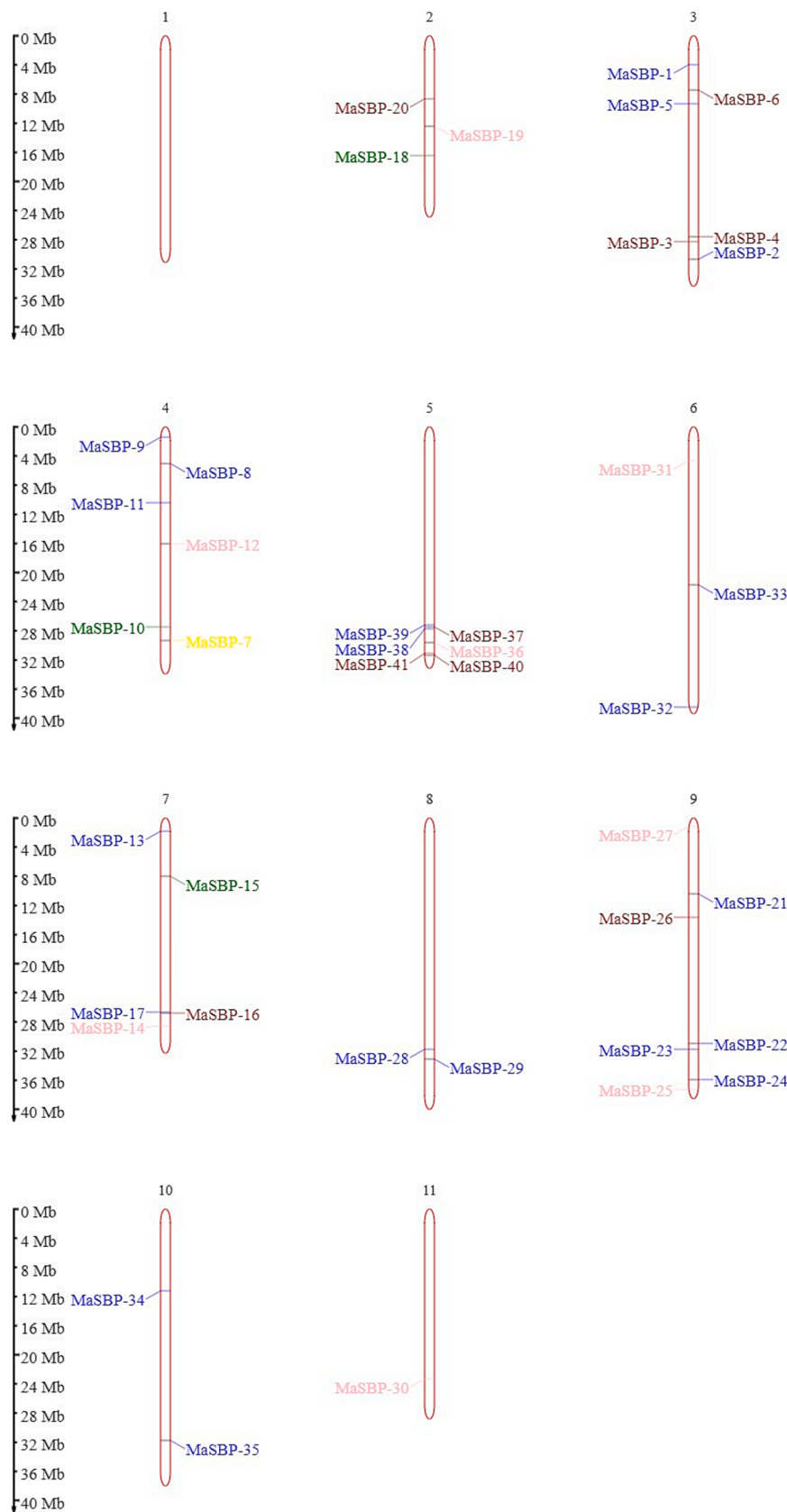


Fig. 7. Chromosomal location map. The scale indicates chromosomal distances of 40 Mb. Above them, the chromosomal numbers are indicated.

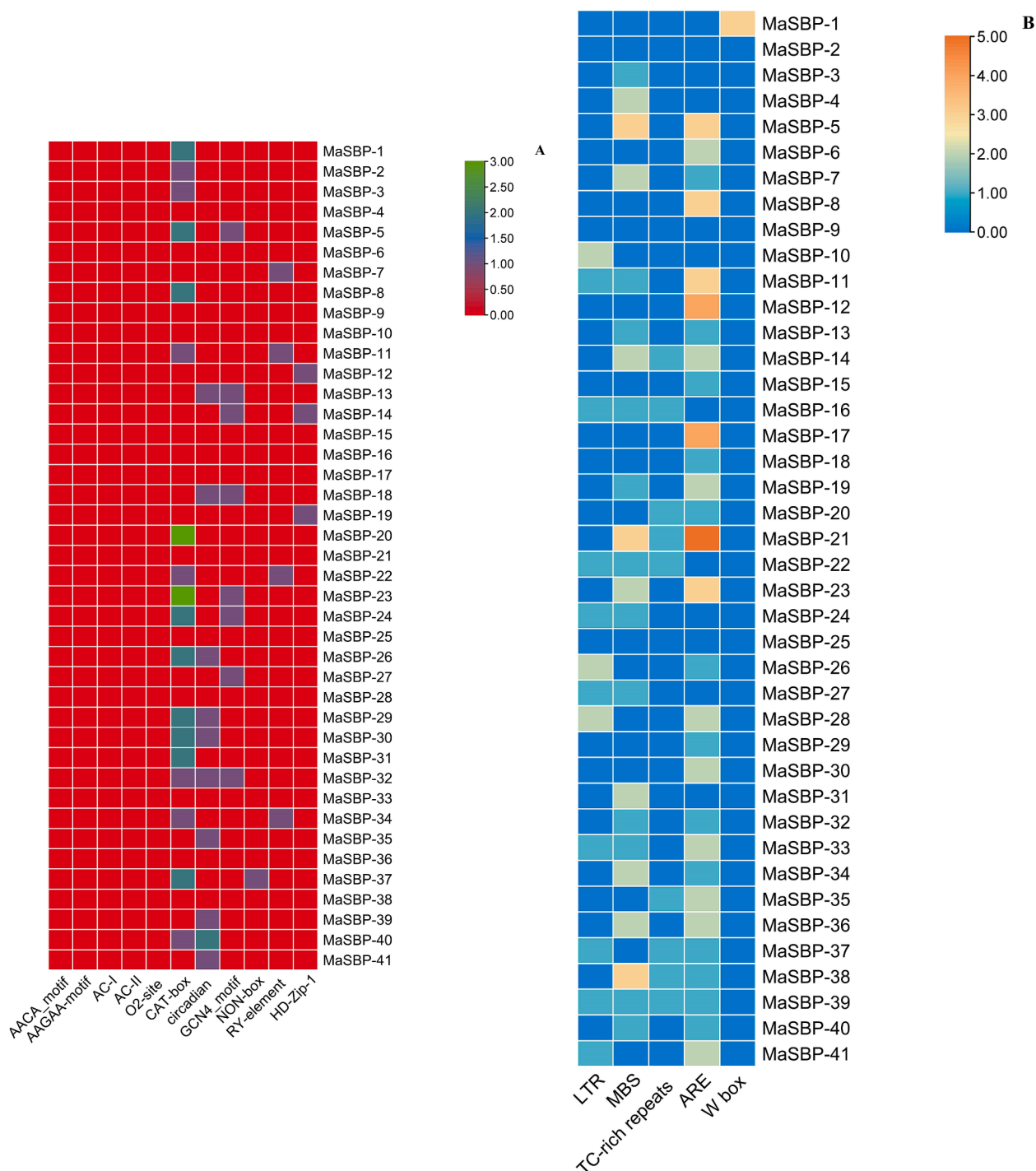


Fig. 8. Heatmap of fundamental *cis*-elements in *MaSBPs*. Six types of *cis*-elements: (A) Development, (B) Environmental stress (C) Hormone, (D) Site-binding, (E) Light responsive, and (F) Transcription related elements were identified in the promoter regions. The number of *cis*-elements is represented by different colors.

additional data imported from the Gene Expression Omnibus (GEO) at NCBI⁵³. To identify the *SBP* genes in *M. acuminata* that may play a role in regulating specific organ growth, development, and stress tolerance, the expression levels of *MaSBPs* were examined. Atlas database contained information about a total of 3 experiments (baseline:1; differential:2), that were conducted in *M. acuminata*. Except for a few (*MaSBP*-7, *MaSBP*-10, *MaSBP*-18), nearly all *MaSBP* genes (especially *MaSBP*-3, *MaSBP*-20, *MaSBP*-37, *MaSBP*-40) were expressed in the endocarp organ portion during the fruit development stage in the baseline expression settings (Fig. 12). Regarding *M. acuminata*, data from two differential

experiments were recorded in Expression Atlas, that are (1) RNA-Seq of banana plants in response to *Bacillus amyloliquefaciens* and *Pseudomonas fluorescens*, (2) RNA-Seq of banana plants in vitro roots under mild osmotic stress. Six genes (*MaSBP*-4, *MaSBP*-5, *MaSBP*-23, *MaSBP*-28, *MaSBP*-38, *MaSBP*-39) and one gene (*MaSBP*-20) out of all forty-one *MaSBPs* were expressed in root, pseudostem and leaf tissues in response to *B. amyloliquefaciens* and *P. fluorescens* infection respectively (Fig. 13). The differential expression unit $\log_2$ -fold change evaluates the difference in gene expression levels between two situations. A positive Log2FC value indicates that these genes are upregulated or more highly

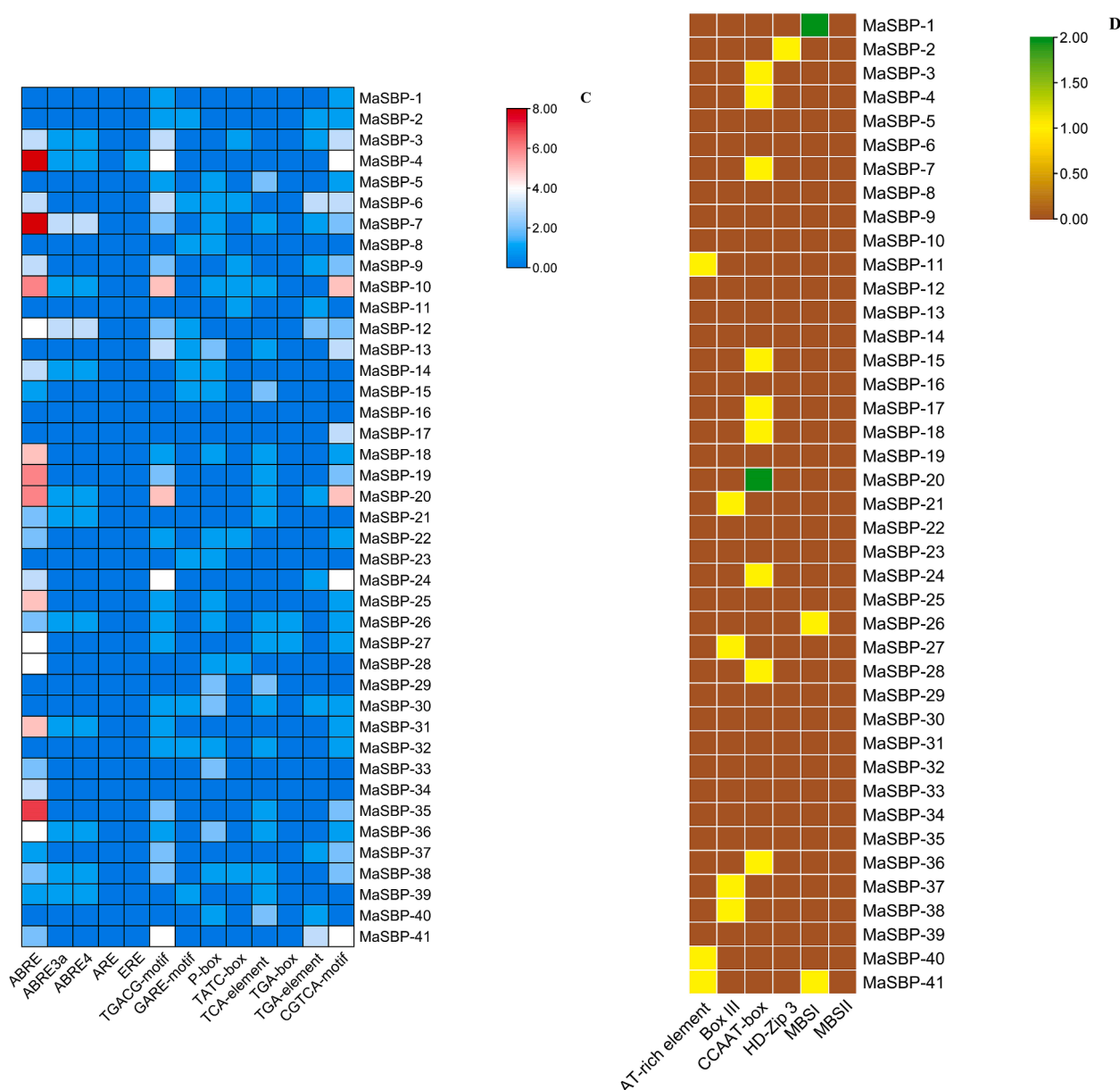


Fig. 8. (continued).

expressed in the test group. However, the expressions of *MaSBP-20* and *MaSBP-23* were downregulated, as shown by a negative Log2FC value. The test sample's roots exhibited reduced expression of the *MaSBP-7* gene during osmotic stress (Fig. 13).

4. Discussion

SQUAMOSA promoter binding protein (SBP) genes code for a group of transcription factors that are unique to plants and play roles in flower and fruit development, leaf initiation, phase transitions, and embryonic development⁵⁴. While the SBP gene family has been identified in various plant species, a thorough analysis of this gene family in banana, a monocot plant, has not yet been performed. In this study, the genome-wide analysis found 41 SBP genes in *M. acuminata*, which were labelled *MaSBP-1* to *MaSBP-41*. *M. acuminata* has 41 SBP genes, which is roughly twice the number found in Arabidopsis and rice, but similar to that of other monocots like wheat (48)⁵⁵, maize (31)⁶, and sugarcane (30)⁵⁴. This indicates that SBP genes in different monocot and dicot species have undergone distinct gene duplication processes⁵⁴. A

phylogenetic study was conducted to compare 31 SBP protein sequences from dicot Arabidopsis and monocot rice with 41 *MaSBPs* and the result was categorized into five groups (G1-G5) based on sequence homology. Moreover, the tree includes both orthologous and paralogous sequences. "Paralogs" are genes that have diverged within the same organism as a result of gene duplication. While these newly formed genes typically share characteristics with their ancestral genes, they take on different functions⁵⁶.

All *MaSBP* gene protein products were found to contain a highly conserved SBP domain, as confirmed by domain analysis. Multiple sequence alignment showed that each *MaSBP* includes two zinc finger motifs (C3H and C2HC types) and one overlapping nuclear localization signal (NLS) region within the SBP domain, which are key distinguishing features of SBP-box proteins⁵⁵. A nuclear localization signal (NLS) is a short amino acid sequence that aids in the transport of nuclear proteins into the nucleus. NLS motifs are significant for two reasons: (1) removing the NLS typically blocks nuclear import, and (2) when a non-nuclear protein is attached to an NLS, it is often successfully imported into the nucleus^{57,58}.

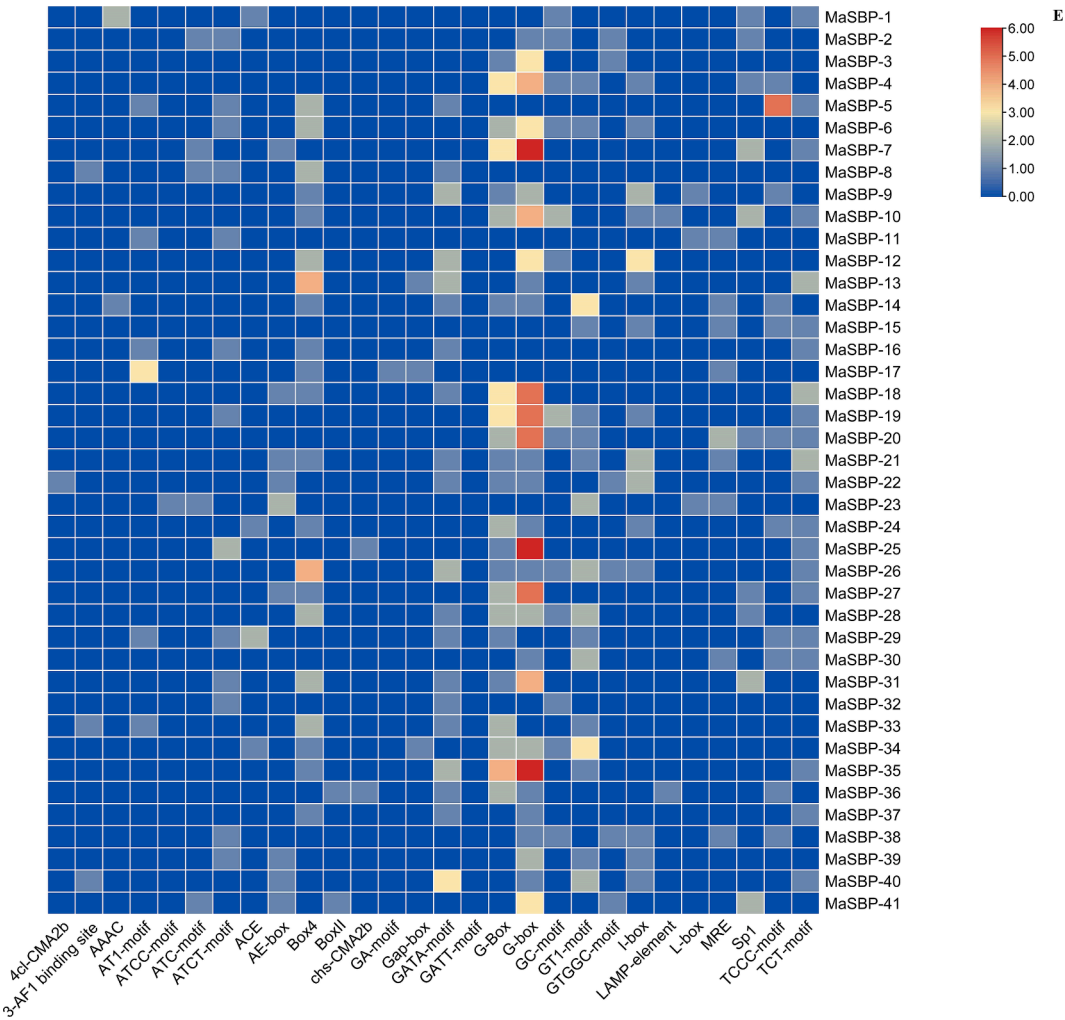


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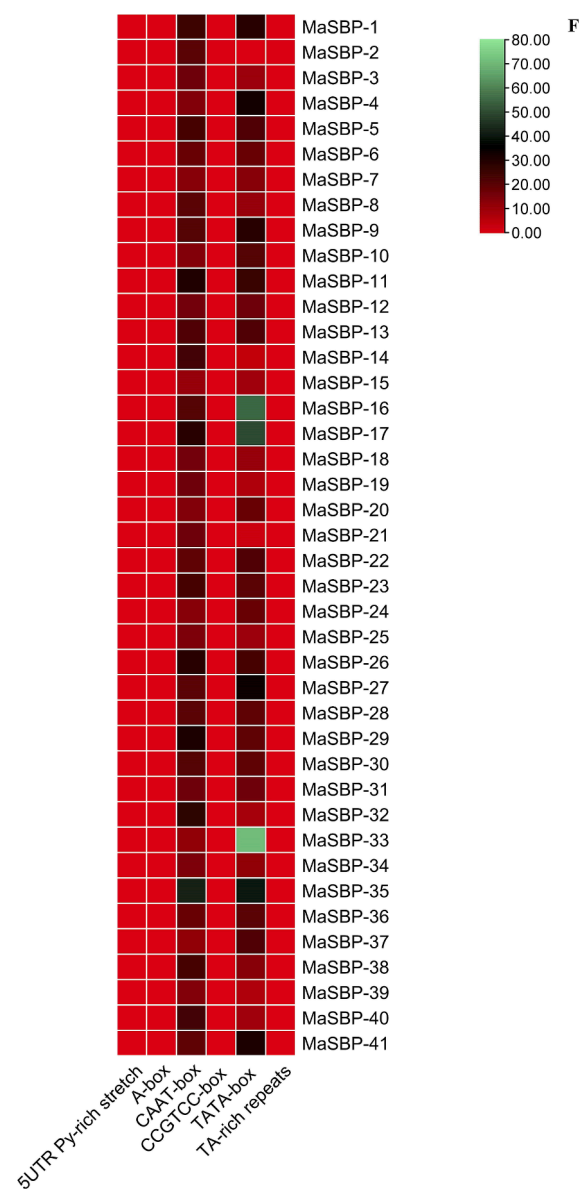


Fig. 8. (continued).

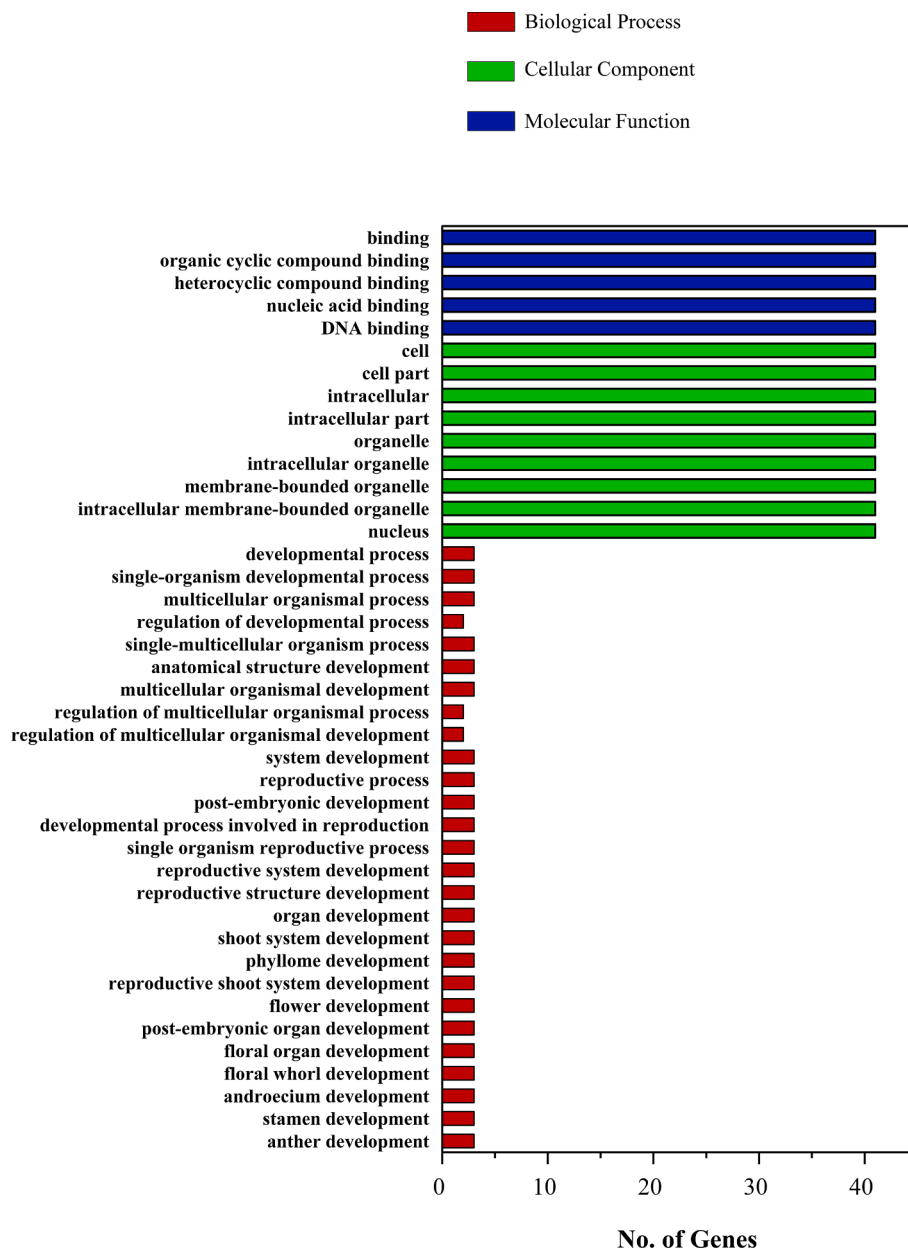
Fig. 9. Gene ontology (GO) graph of *MaSBPs*.

Table 3

Ka/Ks ratios and estimated divergence times for the duplicated *MaSBP* genes.

Paralogous gene pairs		Ka	Ks	Ka/Ks	Time (MYA)
<i>MaSBP-28</i>	<i>MaSBP-34</i>	0.26707108	0.569073391	0.469308677	63.23037678
<i>MaSBP-8</i>	<i>MaSBP-11</i>	0.116788873	0.430440835	0.27132387	47.82675946
<i>MaSBP-32</i>	<i>MaSBP-38</i>	0.186016948	0.440839999	0.421960232	48.98222208
<i>MaSBP-35</i>	<i>MaSBP-39</i>	0.239396382	0.476121758	0.502804962	52.90241751
<i>MaSBP-17</i>	<i>MaSBP-33</i>	0.195051039	0.679839212	0.286907604	75.53769024
<i>MaSBP-9</i>	<i>MaSBP-13</i>	0.210779069	0.840402591	0.250807257	93.37806565
<i>MaSBP-24</i>	<i>MaSBP-29</i>	0.12443214	0.381573888	0.326102347	42.39709862
<i>MaSBP-12</i>	<i>MaSBP-14</i>	0.383263603	0.982041888	0.390272154	109.1157654
<i>MaSBP-25</i>	<i>MaSBP-27</i>	0.223020325	0.714600218	0.31209104	79.40002421
<i>MaSBP-16</i>	<i>MaSBP-37</i>	0.19847731	0.633165701	0.313468196	70.35174457
<i>MaSBP-3</i>	<i>MaSBP-40</i>	0.148343404	0.514990575	0.288050716	57.22117498
<i>MaSBP-4</i>	<i>MaSBP-41</i>	0.200297361	0.584125744	0.342901101	64.90286039
<i>MaSBP-6</i>	<i>MaSBP-26</i>	0.20937102	0.526664848	0.397541284	58.51831642
<i>MaSBP-10</i>	<i>MaSBP-18</i>	0.173512405	0.769634402	0.225447829	85.51493358

*Ka: non-synonymous substitution rate, Ks: synonymous substitution rate, Ka/Ks: selection pressure ratio.

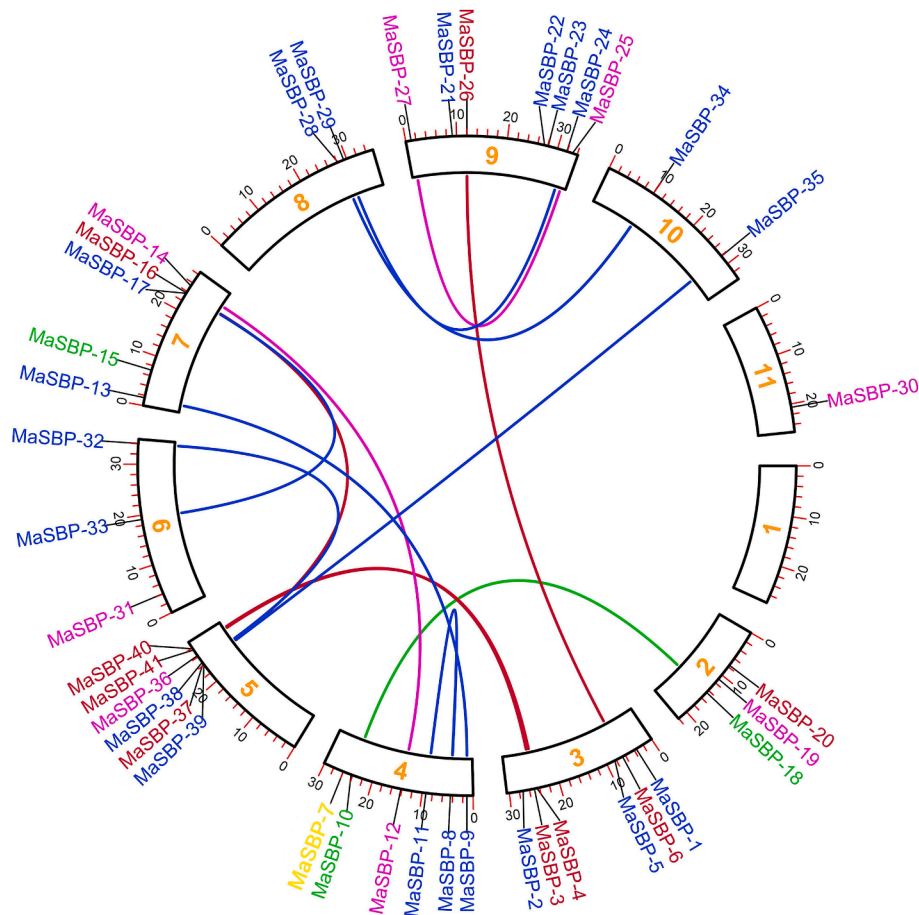


Fig. 10. Circos plot showing duplication events of MaSBPs.

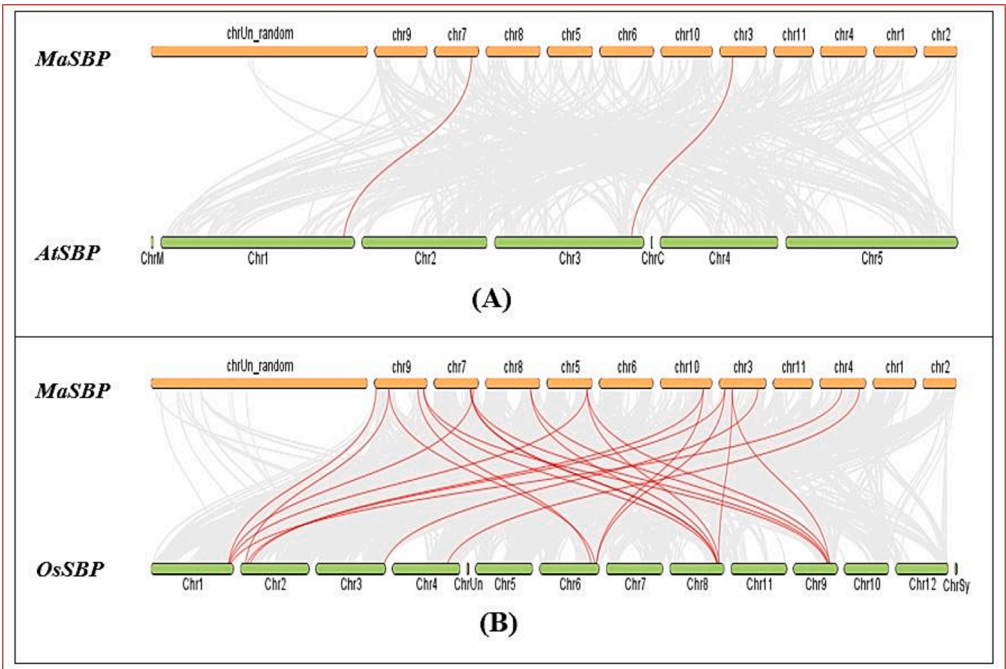


Fig. 11. Synteny graph of AtSBP-MaSBP-OsSBP genes. (A) Banana-Arabidopsis and (B) Banana-Rice homologous gene pairs. The gray lines in the background illustrate the collinear blocks within the genomes of banana, Arabidopsis, and rice, while homologous gene pairs are highlighted with red lines.

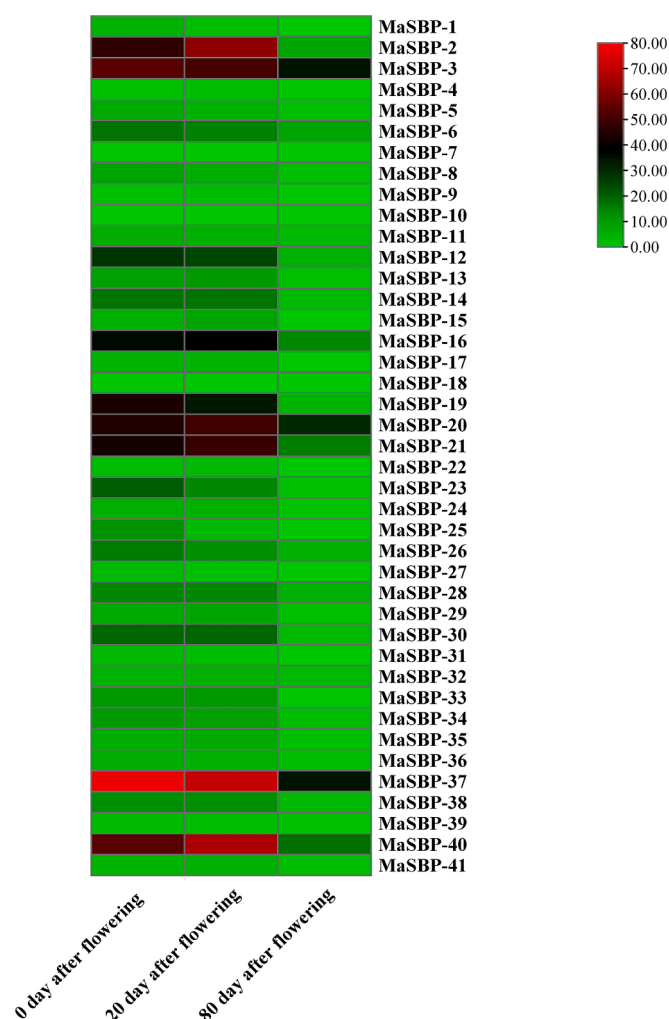


Fig. 12. Heatmap of baseline expression of *MaSBP* genes. The levels of gene expression during fruit development are indicated by different colors: red signifies high expression, black represents moderate expression, and green denotes low expression.

The ExPASy ProtParam program was then used to analyze different protein attributes, and it projected that the MWs and amino acid (aa) lengths of the *MaSBP* proteins varied greatly. Similarly, the theoretical isoelectric points (pI) of these identified proteins were more than 7, with the exception of seven proteins (*MaSBP-3*, *MaSBP-4*, *MaSBP-19*, *MaSBP-20*, *MaSBP-23*, *MaSBP-26*, and *MaSBP-40*), indicating that the majority of the proteins are basic because compounds with $pI > 7$ are considered basic, and those with $pI < 7$ are considered acidic⁵⁹. These proteins' high aliphatic index implies that they are thermostable across a wide temperature range⁶⁰. Furthermore, all of these proteins are hydrophilic since they have a negative GRAVY score⁶¹. The chromosomal distribution of *MaSBP* genes revealed that all genes are spread across ten out of the twelve chromosomes of *M. acuminata*.

The *SPL* family in the monocot plant maize exhibited a diverse range of exon distributions based on gene structure analysis. The arrangement of most maize *SPLs* within the same subfamilies on the evolutionary tree was consistent. A total of 20 conserved motifs were identified in maize *SPL* proteins. Genes belonging to the same subfamily displayed similar motif compositions when analyzed phylogenetically⁶. In case of *MaSBP* sequences, they had varying numbers of exons and introns. There was a total of ten different structural motifs predicted, with the first two motifs being present in all the detected proteins and representing the conserved *SBP* domain. *MaSBP* genes of same group have almost similar motif composition and structure, which may represent their evolutionary

connections with functional conservation.

The transcription factors are mostly found in the nucleus⁶². Based on subcellular localization findings, nearly all *MaSBP* are found inside the nucleus. In wheat, the promoters of *TaSBP-box* genes contained four types of *cis*-acting elements: those associated with transcription, development, hormones, and responses to abiotic stress⁵⁵. Bioinformatics analysis revealed that *MaSBP* promoter region also contains several *cis*-acting associated with light, hormones, stress, site binding, development, and transcription. Several of these *cis*-elements, including ARE, MBS, LTR, and CAT-box, which regulate plant growth, development, and responses to environmental factors, were found to be abundant in the *MaSBPs*. Notably, only *MaSBP-1* contained three W-box elements associated with wound and pathogen responses. Following this, phytohormone-regulating elements were present, with signaling molecule elements like ABRE, TGACG-motif, TATC-box, and CGTCA-motif being the most prominent, playing critical roles in abscisic acid and MeJA responsiveness. Mostly G-box and TCCC motifs were detected in putative *MaSBP* sequences in response to light. The regulatory elements of the *MaSBPs* promoter were preserved, as they were in other model plants.

The clustering of *SBP* gene family members with similar structures and functions suggests a shared evolutionary origin. This indicates that gene duplication events are crucial for gene expansion and evolution, leading to the formation of multiple paralogous gene pairs that help plants adapt to various developmental pathways⁵⁴. A genome-wide investigation of monocot fruit, sugarcane (*Saccharum spontaneum*) revealed a total of 30 *SBP* genes. According to phylogenetic study, these genes have orthologous links with *SBP* members from Arabidopsis and rice. *SsSBPs* had a stronger interaction with *SBP* proteins from *O. sativa*, compared to Arabidopsis. The Ka/Ks ratios of sugarcane *SBP* genes were below 1, indicating that these genes underwent strong purifying selection to remove harmful mutations after duplication during evolution⁵⁴. Similar findings were observed in the case of the banana plant. Synteny analysis revealed two *SBP* orthologous pairings between the genomes of *M. acuminata-A. thaliana* and twenty-four orthologous pairs were found in *M. acuminata-O. sativa*. It clearly indicates that the dicot Arabidopsis and monocotyledonous plant banana diverged before the division of banana and rice. Phylogenetic data (Fig. 4) showed fourteen pairs of paralogous protein among the 41 *MaSBP* proteins. One formula for estimating gene duplication is the Ka/Ks ratio. The synteny analysis indicated that the Ka/Ks ratio was below one for all duplicated *MaSBP* gene pairs, implying that purifying selection took place after the duplication events.

Genes necessary for growth and development could be found through studying the GO annotation and expression profiles of *SBP* protein sequences in *M. acuminata*. According to the findings of GO analysis, *MaSBP-19*, *MaSBP-21*, *MaSBP-29* might have some contribution in flower organ, fruit formation and shoot system development process. This hypothesis was validated by the baseline expression data, that demonstrated the function of *MaSBPs* during fruit developmental stage. Genes *MaSBP-3*, *MaSBP-20*, *MaSBP-37*, *MaSBP-40* were more expressed among others. These results are comparable to the microRNA research of *SPL* genes in Arabidopsis, which showed that they are essential for governing flowering systems in plants¹². The differential expression analysis of *MaSBPs* offered insights into how genes are regulated in response to specific biotic and abiotic stresses. For instance, *MaSBP-4*, *MaSBP-5*, *MaSBP-23*, *MaSBP-28*, *MaSBP-38*, *MaSBP-39*, *MaSBP-20*, *MaSBP-7* exhibited upregulation and downregulation respectively in response to pathogen infection and osmotic pressure.

5. Conclusion

The current study conducted an in-silico analysis of the *SBP* gene family in *M. acuminata* (banana) at the whole-genome level. There is a total of 41 *MaSBP* genes, which were phylogenetically classified into 5 groups. Gene structure, conserved domain and motif composition, and

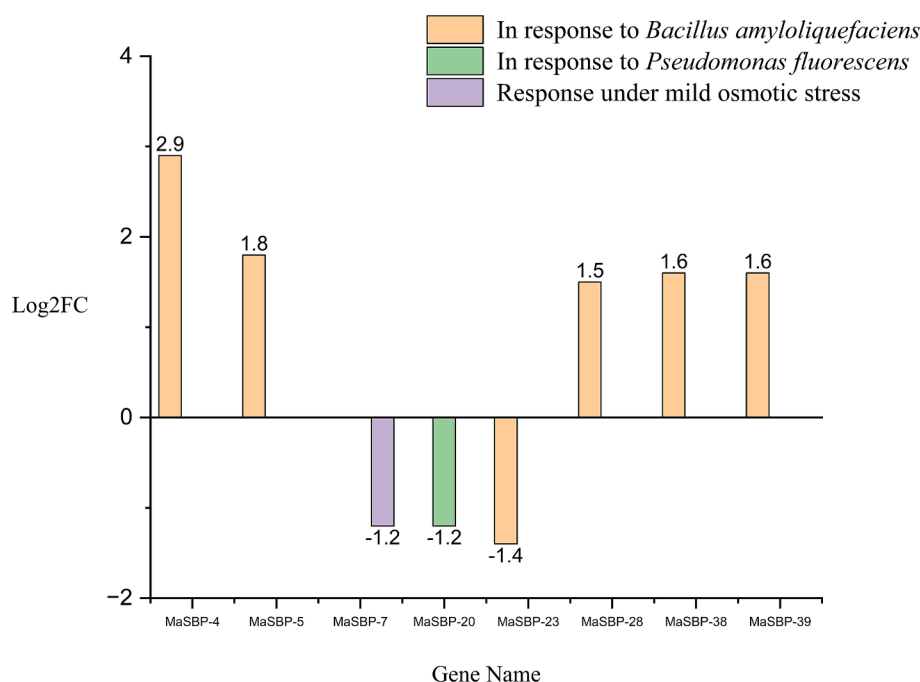


Fig. 13. MaSBPs differential gene expression graph.

other data also supported this conclusion. In-depth research was further aided by the examination of CREs, evolutionary relationship, and divergence. The discovered MaSBP proteins' subcellular localization confirmed their roles as SBP transcription factors. The results of this study can thus be used to explore the functioning and physiological role of the SBP genes found in *M. acuminata* in the future.

Consent for publication

Not Applicable.

Ethics approval and consent to participate

Not Applicable.

Availability of data and materials

All data generated or analyzed during this study are included in this published article [and its [supplementary information files](#)].

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CRedit authorship contribution statement

Israt Jahan Mouri: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Md. Shariful Islam:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Not Applicable.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jgeb.2025.100461>.

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